

Quotation Module Product Requirements Document (PRD)

Comprehensive Audit and Target-State Specification

- Document Version: 1.0
 - Date: 2026-04-07
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 - Authoring Method: Codebase Audit + Existing Technical Artifact Synthesis
 - Scope: Quotation and Quote Composer ecosystem (UI, APIs, services, schema, integrations, governance)
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0) Executive Summary

This PRD is built from a systematic audit of the current Quotation implementation. It documents:

1. Full inventory of modules and dependencies across routes, components, services, APIs, RPCs, edge functions, and data schema.
 2. Section-level behavior for each primary module, including interaction models and conditional logic.
 3. Field-level contract documentation for key entities, including validations, defaults, formulas, and persistence mapping.
 4. General information requirements: role/permission hierarchy, audit trail standards, integration APIs, and compliance.
 5. Detailed Quotation Composer specifications (current + target), with visual and interaction requirements.
 6. Current-state analysis, screenshots, flowcharts, and gap matrixes.
 7. Future-state roadmap and architecture proposals with scalability and resilience controls.
 8. Persona-driven stories with explicit acceptance criteria.
 9. MoSCoW prioritization with estimates and staffing guidance.
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1) Systematic Audit Baseline

1.1 Audited Sources (Primary Evidence)

- UI routes and pages:
 - `src/pages/dashboard/QuoteNew.tsx`
 - `src/pages/dashboard/QuoteDetail.tsx`
 - `src/pages/dashboard/Quotes.tsx`
 - `src/pages/dashboard/QuotesImportExport.tsx`
 - `src/pages/dashboard/QuoteTemplates.tsx`
 - `src/pages/dashboard/QuoteBookingMapper.tsx`
- Core composer and form modules:
 - `src/components/sales/unified-composer/UnifiedQuoteComposer.tsx`
 - `src/components/sales/unified-composer/FormZone.tsx`
 - `src/components/sales/unified-composer/ResultsZone.tsx`
 - `src/components/sales/unified-composer/FinalizeSection.tsx`
 - `src/components/sales/unified-composer/schema.ts`
 - `src/components/sales/quote-form/*`
- Services and domain logic:
 - `src/services/quotation/*`
 - `src/services/QuoteOptionService.ts`
 - `src/services/pricing.service.ts`
- API and integration contracts:
 - `src/pages/api/v2/quotations/import.ts`
 - `src/pages/api/v2/quotations/export.ts`
 - `docs/api/quote-mapping-api.yaml`
- Database contracts and migration evolution:
 - `supabase/migrations/20251005170000_quotation_core.sql`
 - `supabase/migrations/20251108115738_69577924-736f-40dd-9d24-0a104675c62e.sql`
 - `supabase/migrations/20260122090000_enhance_quote_schema.sql`
 - `supabase/migrations/20260201170001_enterprise_quote_architecture.sql`
 - `supabase/migrations/20260215000000_quote_cargo_configurations.sql`
 - `supabase/migrations/20260227000002_enhance_quotation_module.sql`
 - `supabase/migrations/20260303120000_update_save_quote_atomic_locations.sql`
 - `supabase/migrations/20260306180000_update_quote_templates_schema.sql`
- Role and access model:
 - `src/config/permissions.ts`
- Existing analysis and performance evidence:
 - `docs/quote-data-pipeline.md`

- docs/quote-generation-workflow.md
- docs/quick-quote-integration-spec.md
- docs/technical_specs/quotation_engine_architecture.md
- tests/e2e/quotation/quotation-e2e-review-and-traceability.md

1.2 Current Product Posture (As-Is)

- The platform supports end-to-end quotation lifecycle: create, enrich, compare, persist versions, generate PDFs, import/export, and map quote to booking.
- Core persistence centers around quotes, quotation_versions, quotation_version_options, quotation_version_option_legs, quote_charges, and cargo/item extensions.
- Composer already supports complex multimodal option editing and financial recalculation, but collaborative editing and true drag-and-drop layout composition are not fully realized in the quotation creation flow.
- Quote template and PDF-generation ecosystem is mature and schema-driven; visual-template designer capability is partially implemented and partially planned.

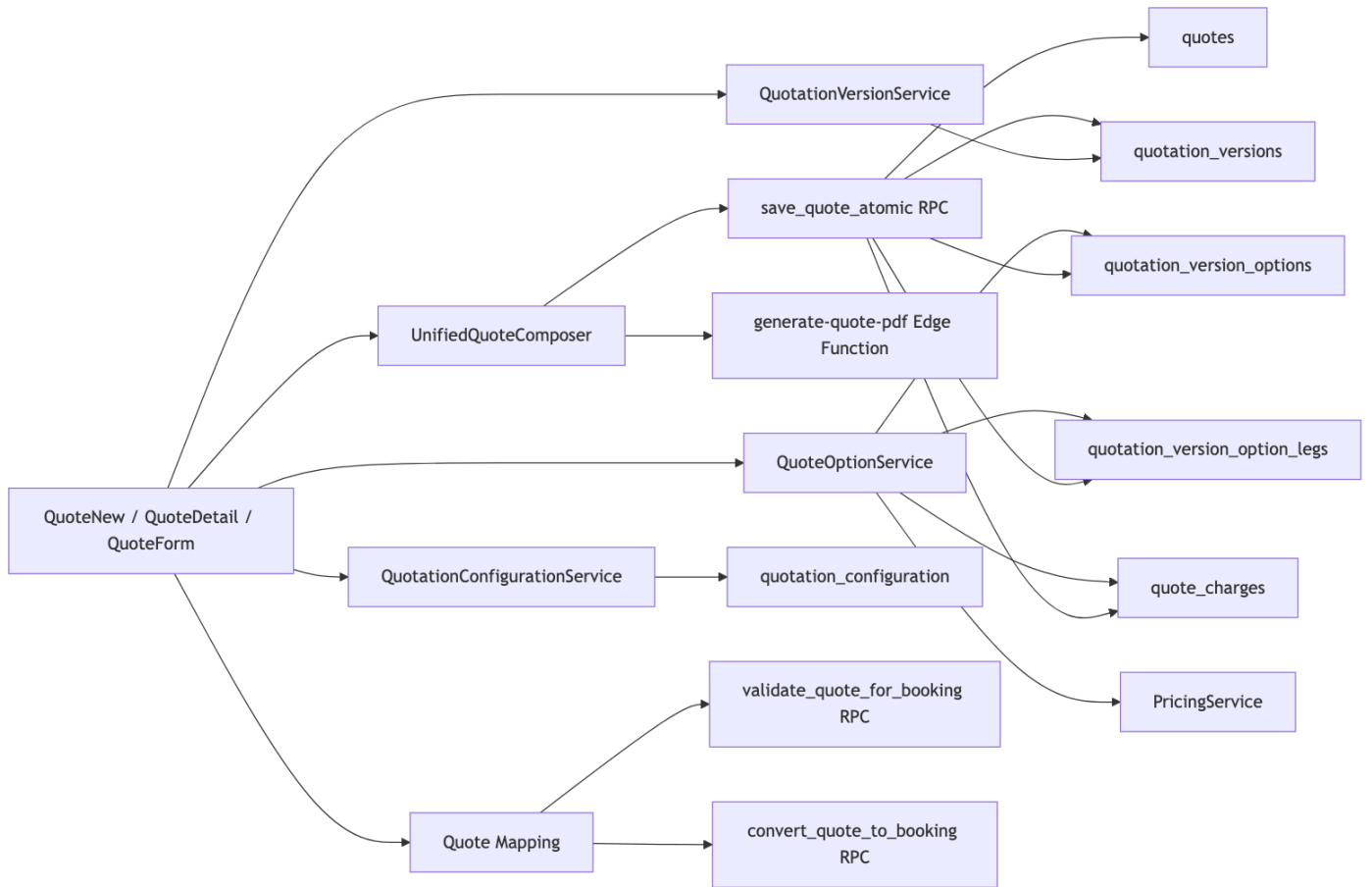
2) Quotation System Architecture Inventory

2.1 Module Inventory

Layer	Module	Responsibility	Key Dependencies	Integration Touchpoints
Route	QuoteNew	New quote orchestration, quick/smart quote conversion, option generation	QuoteFormRefactorer, UnifiedQuoteComposer, QuoteOptionService patterns	Supabase tables, RPC save_quote_atomic, rate payload mapping
Route	QuoteDetail	Version management, save/restore version, detail visualization	Version panels, option cards, comparison components	Edge functions save-quotation-version, restore-quotation-version
Route	Quotes	Quote list/search/filter actions	data hooks, table/list components	CRUD endpoints and direct data reads
Route	QuotesImportExport	Batch import/export entry UI	API routes v2, permission checks	/api/v2/quotations/import, /api/v2/quotations/export
Route	QuoteTemplates	Template list/editor and quote creation from template	template components + auth context	quote_templates, quotes insert
Route	QuoteBookingMapper	Quote-to-booking conversion workflow	QuoteSelectionGrid, QuoteMappingPreview	RPC validate_quote_for_booking, convert_quote_to_booking
Component	UnifiedQuoteComposer	Main composer shell (tabs/phases), save, calculate, generate PDF	FormZone, ResultsZone, FinalizeSection, schema	RPC save_quote_atomic, Edge generate-quote-pdf
Component	QuoteFormRefactorer	Structured quote editor (header/logistics/cargo/financials)	QuoteHeader, QuoteLogistics, QuoteLineItems, QuoteFinancials	useQuoteRepository, scoped DB writes
Service	QuoteOptionService	Option, leg, charge normalization and persistence	PricingService, mappers, charge matching utils	quotation_version_options, quotation_version_option_legs, quote_charges

Layer	Module	Responsibility	Key Dependencies	Integration Touchpoints
Service	QuotationVersionService	Create/list/set active/restore logic	DB wrapper and version policies	quotation_versions, quotes.current_version_id
Service	QuotationOptionCrudService	Option-level delete/update wrappers	DB RPC abstraction	RPC delete_quote_option_safe
Service	QuotationConfigurationService	Tenant-level composer defaults and smart mode controls	tenant context	quotation_configuration
Service	QuotationRankingService	Option scoring and recommendation	scoring criteria and metadata	rank fields in quotation_version_options
Service	PricingService	margin rules, financial calculations, realtime subscriptions	margin_rules, service_pricing_tiers, services	RPC calculate_service_price, client fallback
DB Core	quotes + related	Header-level quote contract	contacts/accounts/service types/currency	CRUD and version linkage
DB Versioning	quotation_versions	immutable-ish state progression and snapshots	quote FK, template snapshot	save/restore workflows
DB Options	quotation_version_options	carrier options and ranking metadata	version FK + carrier/reference data	compare/rank/select flows
DB Legs	quotation_version_option_legs	multimodal route legs and transit detail	ports/carrier/service type FKs	per-leg charge mapping and UI route breakdown
DB Charges	quote_charges	buy/sell financial lines	bases/categories/currency/sides	final totals, reconciliation, margin
DB Cargo	quote_cargo_configurations	structured cargo setup by transport mode	package and dimensional metadata	pricing, operational planning
DB Governance	quote_approval_rules, quote_approvals, quote_pricing_logs, quote_events	approvals + auditability	user/tenant context	compliance, enterprise controls

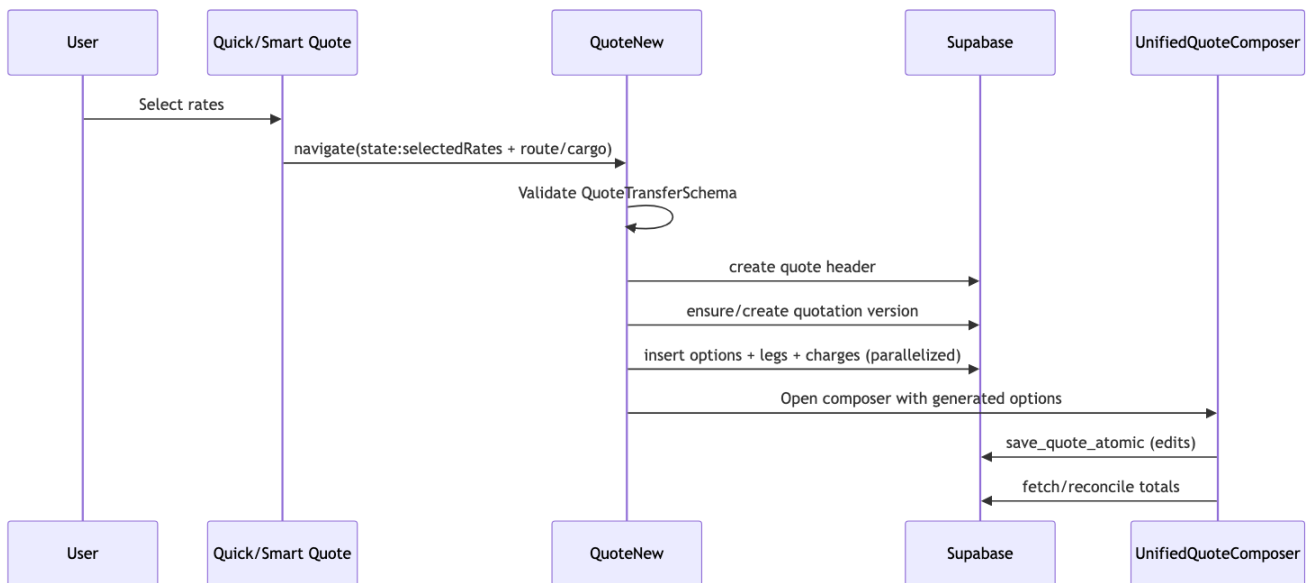
2.2 Dependency Diagram



Mermaid Diagram 1

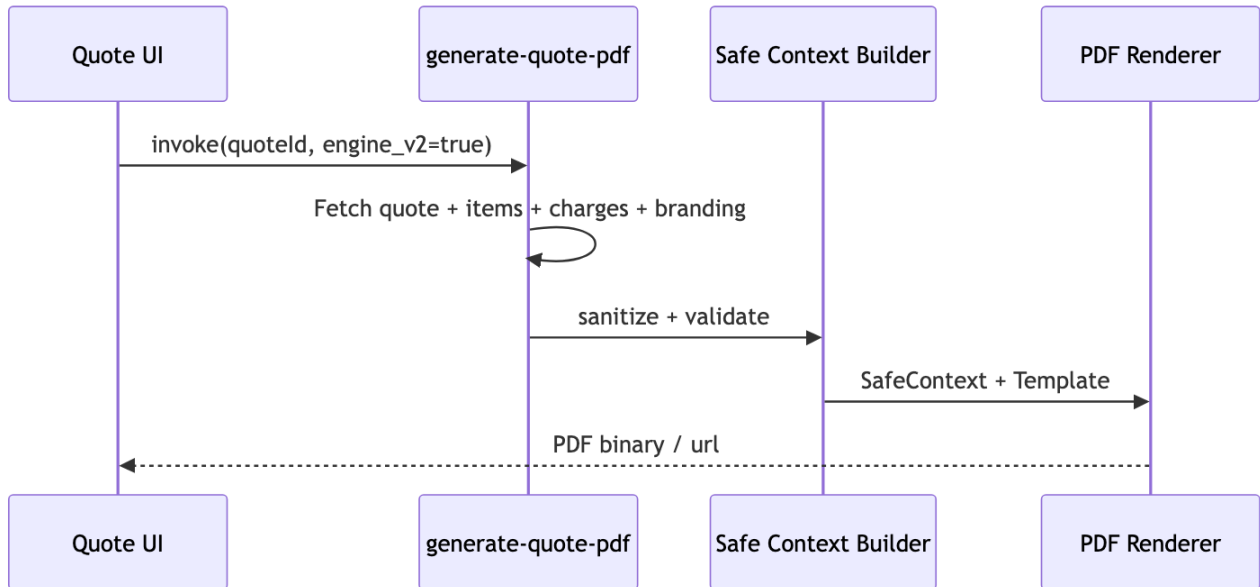
2.3 Data Flow Diagrams

A) Quick/Smart Quote to Composer



Mermaid Diagram 2

B) PDF Generation



Mermaid Diagram 3

3) Section-Level Functional Decomposition

3.1 QuoteNew Primary Sections

Section	Purpose	User Interactions	Business Logic	Conditional Rendering	Performance Expectations
Initialization	Resolve navigation payload and startup data	enter page / convert flow	validates transfer payload, creates quote header/version	shows failure UI when tenant/version missing	initial load under 1s for standard form
Generation Progress	show conversion from selected rates	view progress and errors	parallel option insertion and timeout handling	visible only when selected rates exist	4 options target ~1-3s
Composer Launch	route to detailed editing	edit options / legs / charges	binds generated data to composer state	appears once version/options exist	smooth transition without hard reload

3.2 UnifiedQuoteComposer Sections

Section	Purpose	User Interactions	Business Logic	Conditional Rendering	Performance Requirements
General Information tab	CRM/guest context and core quote header	select opp/account/contact, standalone toggle	cross-field synchronization and validation	guest fields only in standalone	under 100ms field-response
FormZone	route and cargo intake	choose mode, origin/destination, cargo, attachments	mode-specific required fields and schema validation	air/ocean/rail-specific controls	autocomplete lookup <300ms perceived
ResultsZone	option review and editing	compare options, adjust charges/margins	ranking, charge aggregation, smart suggestions	only after rates/options available	list/grid rendering should stay <16ms/frame
FinalizeSection	save/version/pdf actions	save draft/final, generate pdf	payload normalization then save_quote_atomic	enabled when form validity passes	save RPC robust retry and clear error taxonomy

Section	Purpose	User Interactions	Business Logic	Conditional Rendering	Performance Requirements
Error handling surface	resilience to RPC/schema mismatches	user retries with guidance	maps backend error signatures to user-readable remediation	shown only on failure	no hard crash; recoverable fallback path

3.3 QuoteFormRefactored Sections

Section	Purpose	User Interactions	Business Logic	Conditional Rendering	Performance Requirements
QuoteHeader	quote identity + CRM linkage	set title/status/dates and link entities	auto-populate account/contact from opportunity mode	standalone guest block toggled	reactive updates with no full re-render
QuoteLogistics	service/carrier/route configuration	select service type/level/carrier /ports	normalization + leg-derived placeholders	multi-leg rows only when legs exist	lookup + select under 200ms
QuoteLineItems	cargo/commercial line management	add/remove/edit items	maps nested cargo object to flat form state	empty state card when no items	virtualization for large item lists
QuoteFinancials	pricing, taxes, terms, notes	calculate estimate, apply totals, verify server	margin-rule application + reconciliation	warning only when items exist and total is zero	calculation feedback <2s target

3.4 Versioning/Approval/Template Sections

Module	Sections	Notes
quotation-versions/*	history, comparison, actions, PDF generation, approval workflow	supports save/restore and enterprise governance controls
QuoteTemplates	list, editor, selection/create-from-template	schema-backed template content with quote instantiation
QuoteBookingMapper	selection and preview/validation	quote conversion guardrails and audit mapping

4) Comprehensive Field-Level Documentation

Note: Length semantics are inferred from schema and DB type. For text columns, practical limits are product-defined rather than DB hard caps.

4.1 Quote Header Contract (quotes + form schema)

Field	Type / Length	Required	Default	Validation / Rule	Formula / Dependency	DB Mapping
title	string, variable text	Yes	none	min length 1	base identity field	quotes.title
description	string text	No	null	none	optional context note	quotes.description
quote_number	string <= 32 (composer path)	Optional/manual	auto generated when blank	regex [A-Za-z0-9._-]*	uniqueness in sequence policy	quotes.quote_number
status	enum-like text	Yes	draft	UI set (draft/sent/accepted/rejected)	drives available actions	quotes.status
account_id	UUID	Conditional	null	required unless standalone or	syncs contact/opportunity filtering	quotes.account_id

Field	Type / Length	Required	Default	Validation / Rule	Formula / Dependency	DB Mapping
contact_id	UUID	No	null	opportunity used must align to account if account selected	auto-filled from opportunity/contact	quotes.contact_id
opportunity_id	UUID	Conditional	null	required in non-standalone when account absent	backfills account/contact	quotes.opportunity_id
service_type_id	UUID	No	null	selected from service types	filters service list and carrier mode	quotes.service_type_id
service_id	UUID	No	null	must belong to selected service type	dependency on service_type_id	quotes.service_id
incoterms	string	No	null	selected from allowed incoterms list	impacts pricing/compliance context	quotes.incoterms
currency_id	UUID	No	null	selected from currency master	used in charge and totals display	quotes.currency_id
origin_port_id	UUID	No	null	selected from locations	used by routing and leg derivation	quotes.origin_port_id
destination_port_id	UUID	No	null	selected from locations	used by routing and leg derivation	quotes.destination_port_id
pickup_date	date/string	No	null	date input; downstream chronology checks	paired with delivery deadline	quotes.pickup_date
delivery_deadline	date/string	No	null	optional date	should be >= pickup in ops process	quotes.delivery_deadline
vehicle_type	string	No	null	free text	used in logistics and docs	quotes.vehicle_type
special_handling	string/json	No	null	free text / structured in RPC payload	compliance and execution hints	quotes.special_handling
tax_percent	numeric-string	No	0	non-negative	total = shipping + tax	quotes.tax_percent
shipping_amount	numeric-string	No	0	non-negative	subtotal input for financials	quotes.shipping_amount
terms_conditions	text	No	null	none	shown in outbound quote docs	quotes.terms_conditions
notes	text	No	null	internal use	non-customer-facing	quotes.notes
billing_addresses	jsonb	No	{}	standalone mode requires key fields	used for guest quote mode	quotes.billing_address

Field	Type / Length	Required	Default	Validation / Rule	Formula / Dependency	DB Mapping
shipping_address	jsonb	No	{}	optional	address output in docs	quotes.shipping_address
tenant_id	UUID	Yes	context-derived	must match scoped tenant	multitenant isolation	quotes.tenant_id
franchise_id	UUID	Conditional	context-derived	org hierarchy dependent	franchise partitioning	quotes.franchise_id

4.2 Cargo Item Contract (items -> quote_items_core + extension)

Field	Type	Required	Default	Validation	Formula/Dependency	DB Mapping
type	enum loose/container/unit	Yes	loose	enum constraint	drives container fields requirement	logistics.quote_items_extension.type
product_name	string	Yes	none	min length 1	basis for commodity description	quote_items_core.product_name
description	string	No	null	none	can mirror product name	quote_items_core.description
quantity	number	Yes	1	min 1	line total calc	quote_items_core.quantity
unit_price	number	Yes	0	min 0	line total calc	quote_items_core.unit_price
discount_percent	number	No	0	min 0 max 100	discount amount derivation	quote_items_core.discount_percent
commodity_id	UUID	No	null	optional lookup	commodity master link	quote_items_core.commodity_id
aes_hts_id	UUID	No	null	optional lookup	trade compliance linkage	quote_items_core.aes_hts_id
attributes.weight	number	No	0	non-negative	logistics calculations	logistics.quote_items_extension.weight_kg
attributes.volume	number	No	0	non-negative	logistics calculations	logistics.quote_items_extension.volume_cbm
attributes.length/width/height	number	No	0	non-negative	dimensional weight contexts	logistics.quote_items_extension.attributes
attributes.hs_code	string	No	null	optional	customs mapping	logistics.quote_items_extension.attributes
attributes.hazmat	object	No	null	optional structured	hazmat compliance	logistics.quote_items_extension.attributes
container_type_id	UUID	Conditional	null	required for container type	mode/type dependent	logistics.quote_items_extension.container_type_id
container_size_id	UUID	Conditional	null	required for container type	mode/type dependent	logistics.quote_items_extension.container_size_id

Field	Type	Required	Default	Validation	Formula/Dependency	DB Mapping
						nsion.container_size_id

4.3 Cargo Configuration Contract (quote_cargo_configurations)

Field	Type	Required	Default	Validation	Dependency	DB Mapping
transport_mode	text enum-like	Yes	none	one of ocean/air/road/rail	drives downstream logic	quote_cargo_configurations.transport_mode
cargo_type	text enum-like	Yes	none	one of FCL/LCL/Breakbulk/RoRo	impacts container and pricing logic	quote_cargo_configurations.cargo_type
container_type	text	No	null	mode dependent	linked with type_id	quote_cargo_configurations.container_type
container_size	text	No	null	mode dependent	linked with size_id	quote_cargo_configurations.container_size
container_type_id	UUID	No	null	optional FK	references master	quote_cargo_configurations.container_type_id
container_size_id	UUID	No	null	optional FK	references master	quote_cargo_configurations.container_size_id
quantity	integer	Yes	1	min 1	primary volume multiplier	quote_cargo_configurations.quantity
unit_weight_kg	numeric	No	null	non-negative	pricing + routing weight	quote_cargo_configurations.unit_weight_kg
unit_volume_cbm	numeric	No	null	non-negative	pricing + stowage	quote_cargo_configurations.unit_volume_cbm
length_cm/width_cm/height_cm	numeric	No	null	non-negative	used in OOG/LCL handling	corresponding columns
is_hazardous	boolean	No	false	boolean	enables hazardous fields	quote_cargo_configurations.is_hazardous
hazardous_classes	text	Conditional	null	required if hazardous	hazmat rule dependency	quote_cargo_configurations.hazardous_classes
un_number	text	Conditional	null	required if hazardous	hazmat rule dependency	quote_cargo_configurations.un_number
is_temperature_controlled	boolean	No	false	boolean	enables temperature fields	quote_cargo_configurations.is_temperature_controlled
temperature_min/max	numeric	Conditional	null	numeric range expectation	temp-control dependency	corresponding columns
temperature_unit	text	No	c	expected C/F	temp-control dependency	quote_cargo_configurations.temperature_unit

4.4 Option / Leg / Charge Contract

Entity.Field	Type	Required	Default	Validation	Formula/Dependency	DB Mapping
option.option_text_name	text	No	generated label	none	fallback uses carrier+tier	quotation_version_options.option_name
option.total_amount	numeric	Yes	0	non-negative expected	reconciled from sell charges	quotation_version_options.total_amount
option.total_buy	numeric	Derived	0	computed	sum of buy-side charges	quotation_version_options.total_buy
option.total_sell	numeric	Derived	0	computed	sum of sell-side charges	quotation_version_options.total_sell
option.margin_amount	numeric	Derived	0	computed	total_sell - total_buy	quotation_version_options.margin_amount
option.margin_percentage	numeric	Derived	0	computed	(margin / buy)*100	quotation_version_options.margin_percentage
option.rank_score	numeric	No	null	optional scoring	ranking service driven	quotation_version_options.rank_score
option.is_recommended	boolean	No	false	boolean	ranking heuristic output	quotation_version_options.is_recommended
leg.sort_order	integer	Yes	sequential	>0	ordered route sequence	quotation_version_option_legs.sort_order
leg.transport_mode	text	Yes	derived	expected mode code	affects charge matching + carriers	quotation_version_option_legs.transport_mode
leg.origin_location_id	UUID	No	null	FK optional	resolved from name/port mapping	quotation_version_option_legs.origin_location_id
leg.destination_location_id	UUID	No	null	FK optional	resolved from name/port mapping	quotation_version_option_legs.destination_location_id
leg.transit_time_days	number UI	No	0	non-negative	converted to hours in DB writes	mapped to transit_time_hours
charge.category_id	UUID	Yes	fallback category	must resolve category	keyword/fuzzy mapping fallback chain	quote_charges.category_id
charge.basis_id	UUID	No	per_shipment fallback	mapped via basis/unit	fallback to default basis	quote_charges.basis_id
charge.currency_id	UUID	No	USD mapping	mapped via currency code	fallback to quote currency	quote_charges.currency_id
charge.quantity	numeric	Yes	1	non-negative	amount = rate * quantity	quote_charges.quantity
charge.rate	numeric	Yes	computed	non-negative	pricing service output	quote_charges.rate
charge.amount	numeric	Yes	computed	non-negative	pair buy/sell insertion	quote_charges.amount

Entity.Field	Type	Required	Default	Validation	Formula/Dependency	DB Mapping
charge.charge_side_id	UUID	Yes	n/a	must resolve buy/sell side	mandatory side records check	quote_charges.charge_side_id

4.5 Quotation Configuration and Template Fields

Field	Type	Required	Default	Notes
quotation_configuration.default_module	text	Yes	composer	check: composer/legacy/smart
quotation_configuration.smart_mode_enabled	boolean	No	false	toggles smart features
quotation_configuration.smart_mode_settings	jsonb	No	{}	tenant-scoped tuning
quotation_configuration.multi_option_enabled	boolean	No	true	controls option multiplicity
quotation_configuration.auto_ranking_criteria	jsonb	No	weighted cost/time/reliability	ranking inputs
quote_templates.content	jsonb	Yes	none	template body schema-validated
quote_templates.template_name	text	No	null	named template variants
quote_templates.template_options	jsonb array	No	[]	MGL template extension
quote_templates.transport_modes	jsonb array	No	[]	MGL template extension

5) General Information Requirements

5.1 User Roles and Permission Hierarchy (Quotation-Relevant)

Role	View	Create	Edit	Delete	Import/Export	Sensitive Export	Templates Manage	Notes
platform_admin	Yes	Yes	Yes	Yes	Yes	Yes	Yes	full platform scope
super_admin	Yes	Yes	Yes	Yes	Yes	Yes	Yes	enterprise-wide
tenant_admin	Yes	Yes	Yes	Yes	Yes	tenant export only	Yes	tenant scope
franchise_admin	Yes	Yes	Yes	No	Yes	No	No	franchise scope
user	Yes	Yes	limited	No	Yes	No	No	standard sales operations

Permission source slugs include: quotes.view, quotes.create, quotes.edit, quotes.delete, quotes.import_export, quotes.analytics, quotes.export_sensitive, quotes.templates.manage, import_quotation, export_quotation, export_quotation_sensitive.

5.2 Audit Trail Requirements

Mandatory audit points:

- Quote lifecycle events: create/update/status transitions (quote_events).
- Versioning events: save, restore, active version change (quotation_versions + route actions).

- Pricing provenance: automatic rule application and overrides (quote_pricing_logs).
- Approval traces: request/approve/reject with actor and rationale (quote_approvals).
- Mapping actions: quote-to-booking validation and conversion logs.

5.3 Integration APIs and RPC Matrix

Interface	Type	Purpose
/api/v2/quotations/import	REST API route	bulk import orchestration
/api/v2/quotations/export	REST API route	export with permission and sensitivity controls
save_quote_atomic	PostgreSQL RPC	atomic persistence of quote + items + cargo + options + legs + charges
delete_quote_option_safe	PostgreSQL RPC	safe option deletion with integrity rules
validate_quote_for_booking	PostgreSQL RPC	pre-conversion business rule validation
convert_quote_to_booking	PostgreSQL RPC	quote-to-booking conversion with audit continuity
generate-quote-pdf	Edge Function	schema-safe PDF generation
save-quotation-version / restore-quotation-version	Edge Functions	version snapshots and restore path
calculate-quote-financials	Edge Function	server-side financial verification
calculate_service_price	PostgreSQL RPC	service-tier pricing calculation

5.4 Compliance Standards (Required)

- Multitenant data isolation via tenant_id/franchise_id + RLS.
- Scoped access patterns for all quote data operations.
- Schema-first validation using Zod for form and payload contracts.
- Safe-context PDF rendering to prevent data exfiltration.
- Auditability for pricing decisions and approval workflows.
- Backward-compatible schema evolution and version-aware payload normalization.

6) Quotation Composer: Detailed Specification

6.1 UI/UX Wireframe Specification (Desktop)

+-----+ Header / Breadcrumb / Domain / Actions +-----+	
Tabs: [General Information] [Quotation Composer] +-----+	
Left/Primary (70%)	Right/Secondary (30%)
- FormZone	- Option summary
- Mode / Route / Cargo	- Ranking badges
- Carrier preference	- Totals
- Validation hints	- Save/Version actions
- ResultsZone (option table/cards/list)	
- FinalizeSection (notes/margin/pdf/send)	
+-----+	

6.2 Responsive Requirements

- Desktop (>=1280): full two-zone composition, persistent action rail.
- Tablet (768-1279): stacked sections with sticky action bar.
- Mobile (<768): single-column wizard behavior, collapsible detail groups.
- Accessibility:
 - keyboard navigable section controls;
 - visible focus rings;
 - semantic labels for all select/input/button elements.

6.3 Drag-and-Drop Functional Requirements

Current: - Option and charge editing is primarily form-driven; no complete drag-and-drop layout composer for quote-building sections.

Required: - Drag to reorder legs, charge groups, and optional summary blocks. - Drag constraints: - non-removable mandatory sections pinned; - route continuity validator blocks invalid drops. - Interaction standards: - ghost element + insertion indicator; - keyboard-accessible reorder controls; - undo stack integration.

6.4 Real-Time Calculation Engine Requirements

Current: - Uses `PricingService` with cached margin rules and `calculateFinancials`. - Supports estimate generation and optional server verification.

Required: - deterministic calc pipeline per option: - input normalization; - category/basis/currency resolution; - buy/sell pair synthesis; - discrepancy-balancing logic; - reconciliation writeback. - SLA targets: - single option recalc <200ms client-side; - full save+reconcile <1.5s p95 for <=10 options.

6.5 Version Control and Collaborative Editing

Current: - version save/restore exists; - active/current version semantics supported.

Required: - soft-locking with collaborator presence indicator; - conflict resolution: - field-level last-writer metadata; - diff preview before overwrite; - manual merge assistant for critical financial fields. - timeline model: - draft checkpoints; - published snapshots; - immutable PDF/version pairing.

6.6 Interactive Prototype Specifications (Target)

- Prototype scope:
 - quote creation from scratch;
 - quick quote conversion;
 - option comparison/edit;
 - approval trigger path;
 - PDF generation and booking conversion.
 - Prototype fidelity:
 - stateful controls for conditional fields;
 - simulated async states (loading/error/partial success);
 - keyboard-only path validation.
-

7) Current-State Visual Documentation and Gap Analysis

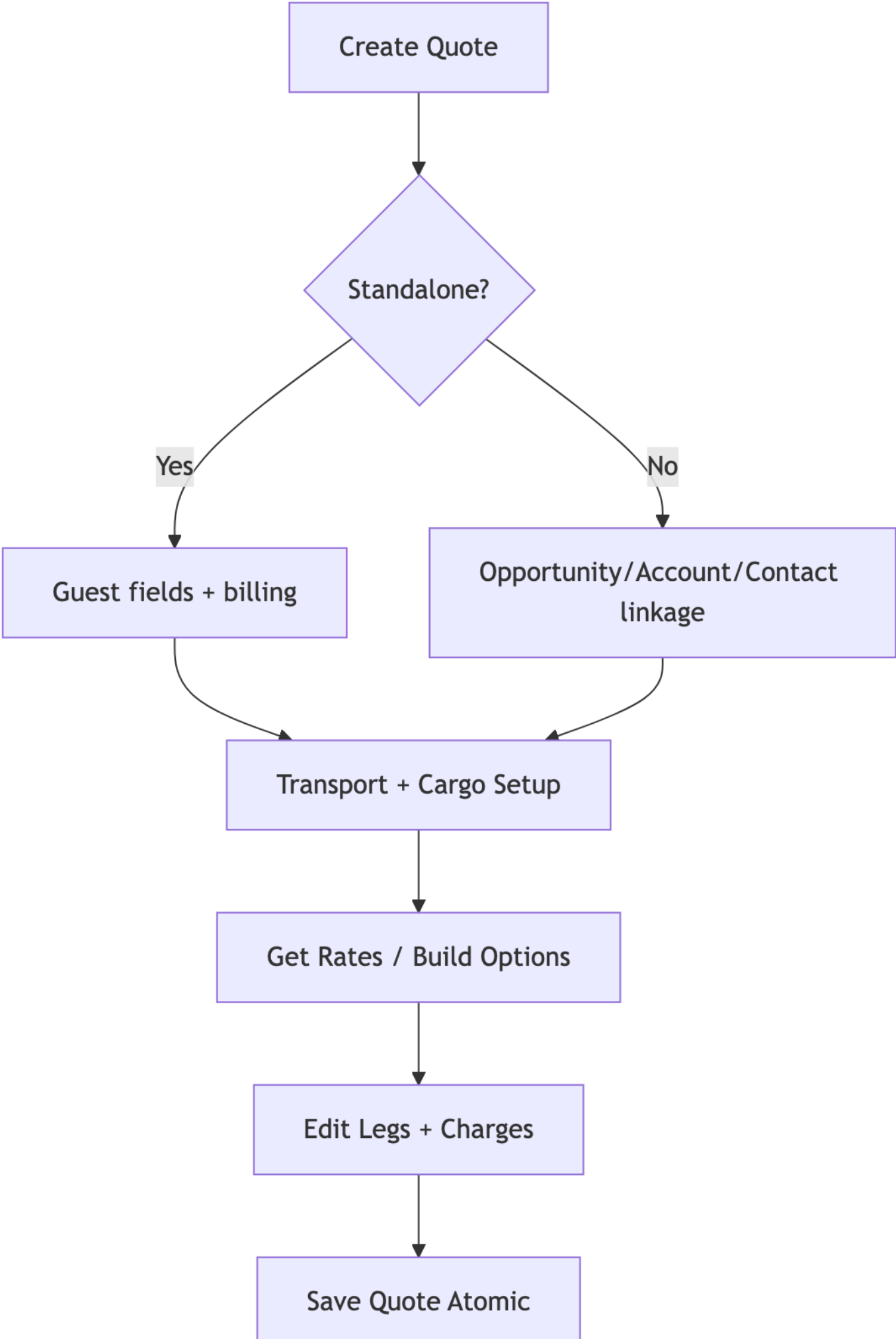
7.1 Screenshot Inventory (Current State)

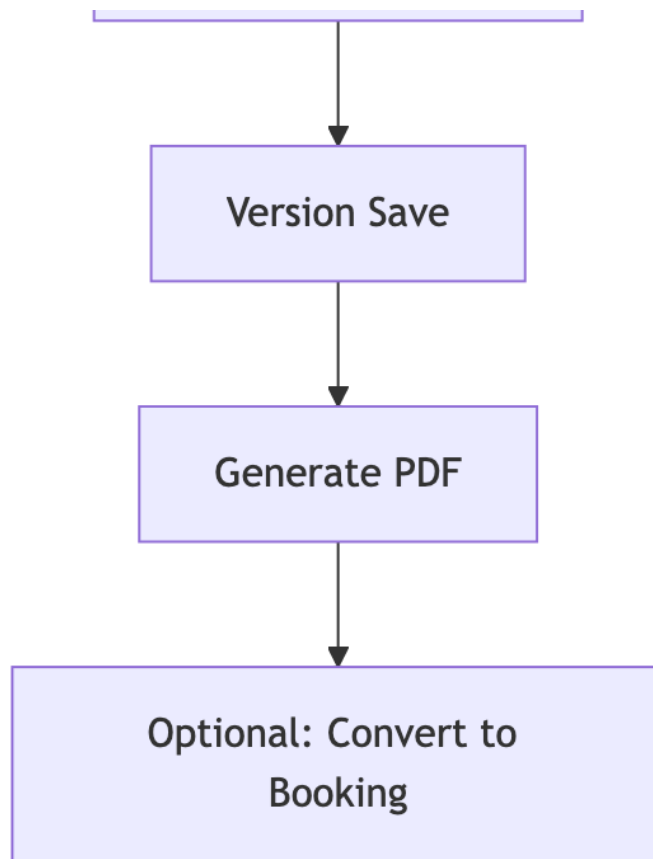
- Desktop snapshot:
 - `tests/e2e/quotation/quotation-comprehensive.spec.ts-snapshots/quotation-composer-visual-chromium-darwin.png`
- Mobile snapshot:
 - `tests/e2e/quotation/quotation-comprehensive.spec.ts-snapshots/quotation-composer-visual-ios-mobile-darwin.png`
- Browser parity snapshots:
 - chromium, chrome-latest, firefox, webkit variants in same folder.

7.2 Annotated Screenshot Legend

1. Header and breadcrumb indicate context and route (`Create Quote`).
2. Top tabset separates General Information and Quotation Composer surfaces.
3. CRM-linked mode toggle controls standalone vs CRM-dependent entry.
4. Context selectors (opportunity/account/contact) drive auto-population logic.
5. Transport mode chips change required cargo constraints and downstream pricing behavior.
6. Commodity and cargo block contains container configuration + dimensions + hazmat controls.
7. Attachments and draft action indicate save pipeline and document input support.

7.3 Current Workflow Diagram (As-Is)





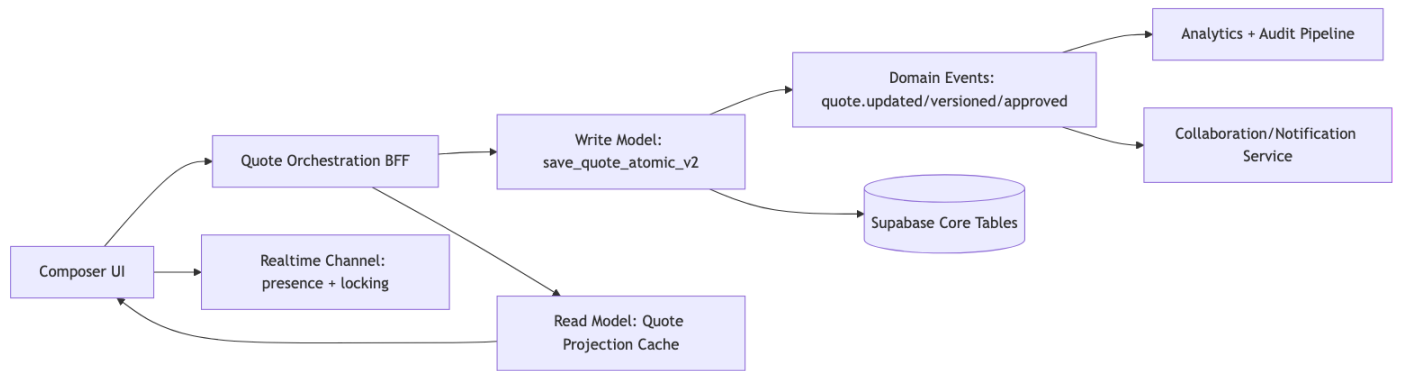
Mermaid Diagram 4

7.4 Gap Analysis Matrix (Existing vs Required)

Capability	Existing State	Required State	Gap Severity	Priority
Section-level drag-and-drop	partial/not universal	full DnD for legs/charges/layout blocks	High	Must
Collaborative editing	none/minimal	realtime presence + conflict handling	High	Must
Field governance registry	split across schemas/migrations	unified machine-readable field catalog	High	Must
Approval orchestration in composer UX	backend tables exist, UX not fully unified	inline approval trigger/escalation flows	Medium	Should
Performance SLO telemetry	partial docs and logs	p95 dashboards by tenant/workflow	Medium	Should
Accessibility automation	basic standards used	WCAG assertion suite + CI gate	Medium	Should
Visual prototype coverage	fragmented	full end-to-end prototype spec and assets	Medium	Should
Predictive ranking explainability	rank fields present	per-option explainability card + trace	Medium	Could

8) Future-State Requirements and Architecture Proposal

8.1 Target Architecture (To-Be)



Mermaid Diagram 5

8.2 Scalability Requirements

- Multi-option save throughput:
 - support 50 options with 5 legs and 40 charges per option under controlled batching.
- Resilience:
 - idempotent save tokens;
 - dead-letter handling for failed async steps (PDF/webhook).
- Data growth:
 - partition or archive strategy for pricing/audit logs.
- Query performance:
 - indexed paths for quote-id, version-id, tenant-id, and status transitions.

8.3 Prioritized Feature Roadmap

Phase	Timeline	Outcomes
Phase 1	4 weeks	unified field dictionary, save flow hardening, error taxonomy, telemetry baseline
Phase 2	6 weeks	collaborative editing MVP, drag-and-drop leg/charge reordering, approval UX integration
Phase 3	5 weeks	projection cache, advanced ranking explainability, accessibility CI and performance gates
Phase 4	4 weeks	enterprise scaling controls, audit dashboards, rollout hardening and migration support

9) User Stories, Personas, and Acceptance Criteria

Persona A: Sales Executive (Fast Quote Creation)

- Story: As a sales executive, I want to convert selected quick quote rates into an editable quotation in one step.
- Acceptance Criteria:
 - selected rates create quote options automatically;
 - progress and failures are visible per-rate;
 - partial success preserves completed options.

Persona B: Pricing Analyst (Charge and Margin Control)

- Story: As a pricing analyst, I need to inspect and adjust buy/sell charges per leg with transparent margin math.
- Acceptance Criteria:
 - per-leg charge edit updates totals instantly;
 - reconciliation detects mismatches >0.01 and flags anomalies;
 - applied margin rules are explainable.

Persona C: Sales Manager (Governance and Approval)

- Story: As a manager, I need low-margin or high-value quotes routed for approval before sending.
- Acceptance Criteria:

- trigger rules execute on save/finalize;
- approval status is visible in quote header and version panels;
- approved/rejected actions are auditable with actor and timestamp.

Persona D: Operations Planner (Route and Cargo Accuracy)

- Story: As operations, I need leg-level route and cargo details to be complete and consistent before booking conversion.
- Acceptance Criteria:
 - origin/destination continuity across legs is enforced;
 - hazardous and temperature fields are validated when applicable;
 - quote-to-booking validation blocks conversion on critical gaps.

Persona E: Tenant Admin (Template and Configuration Governance)

- Story: As tenant admin, I need control over default composer behavior and templates.
- Acceptance Criteria:
 - tenant-specific quotation configuration is editable with RBAC;
 - template versions can be managed without data leakage across tenants;
 - default module behavior (composer/legacy/smart) is applied consistently.

10) MoSCoW Priorities, Estimates, and Resource Requirements

10.1 MoSCoW Matrix

Priority	Feature	Estimate	Roles Needed
Must	unified field registry + schema governance checks	2.5 weeks	1 BE, 1 FE, 1 QA
Must	collaborative edit locking + conflict resolution MVP	4 weeks	1 FE, 1 BE, 1 QA
Must	drag-and-drop leg/charge ordering with validation	3 weeks	2 FE, 1 QA
Must	approval flow integration in composer and detail pages	3 weeks	1 FE, 1 BE, 1 QA
Should	projection cache/read optimization	2 weeks	1 BE
Should	accessibility CI assertions and UX hardening	1.5 weeks	1 FE, 1 QA
Should	performance dashboards (p95 by workflow)	1.5 weeks	1 BE, 1 DevOps
Could	explainable ranking cards and what-if simulation	2 weeks	1 FE, 1 BE
Could	advanced template marketplace workflow	3 weeks	1 FE, 1 BE, 1 PM
Won't (current cycle)	full offline composer mode	deferred	backlog

10.2 Aggregate Delivery Estimate

- Total estimated effort (Must + Should): ~17.5 engineering weeks.
- Suggested squad: 2 FE, 2 BE, 1 QA, 0.5 DevOps, 0.5 PM/BA.
- Suggested rollout: feature-flagged per tenant tier, with pilot tenants before broad release.

11) Visual Documentation Package Requirements (Composer-Focused)

11.1 Annotated Screenshots of Every Interface Element

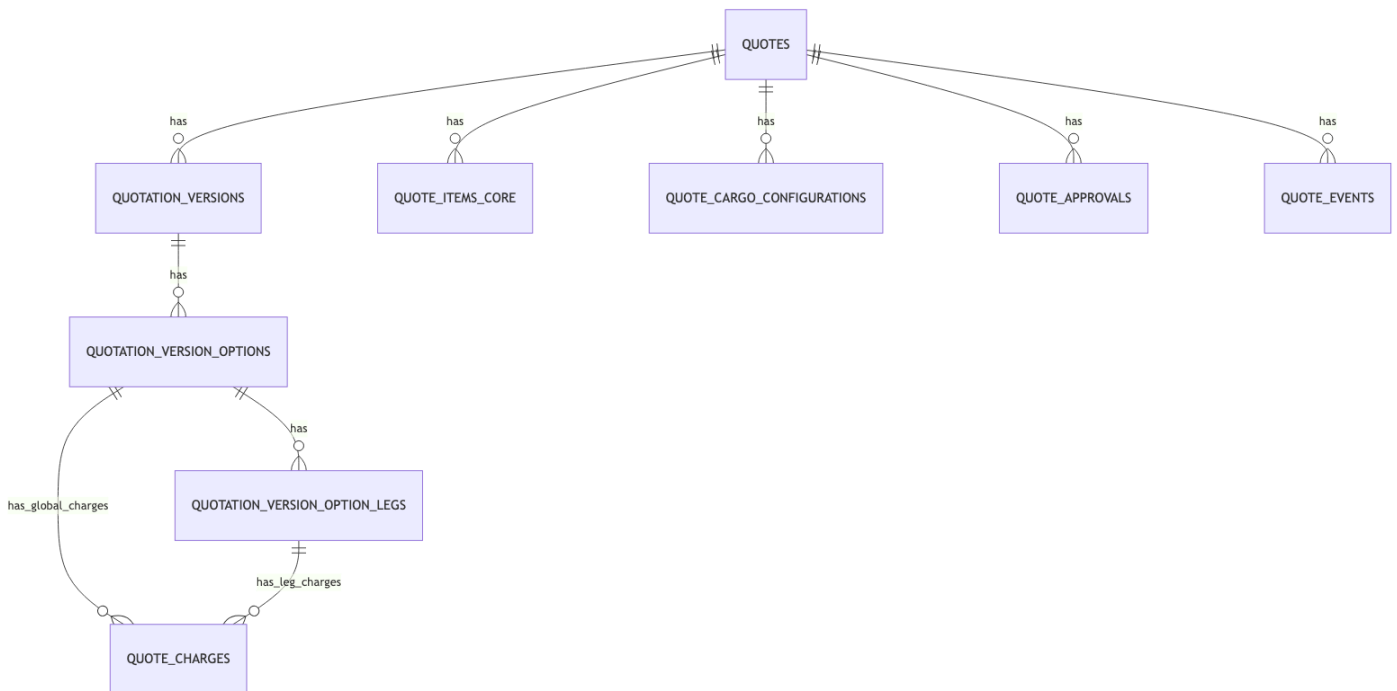
- Required captures:
 - General Information tab (desktop/mobile);
 - FormZone mode variants (air/ocean/road/rail);

- ResultsZone in card/list/grid modes;
 - FinalizeSection save/version/pdf states;
 - error and timeout states.
- Annotation standard:
 - numeric hotspot IDs;
 - field key names;
 - interaction outcome notes.

11.2 Pixel-Perfect Mockups (Module/Section/Row/Column)

- Breakpoints:
 - Desktop 1440x900;
 - Tablet 1024x1366;
 - Mobile 390x844.
- Grid standards:
 - 12-column desktop layout;
 - 8-column tablet;
 - 4-column mobile.
- Tokenized spacing:
 - container padding 24/16/12;
 - section spacing 24;
 - control height 40 standard;
 - card radius 8.

11.3 Comprehensive Field Mapping Diagram



Mermaid Diagram 6

11.4 Interactive Prototype Journey Coverage

- Journey set:
 - create quote from CRM;
 - create standalone quote;
 - convert quick quote to composer;
 - edit options/charges and recalculate;
 - trigger approval and finalize;
 - generate pdf and map to booking.

11.5 Visual Workflow Diagrams

Quotation Creation



General Information



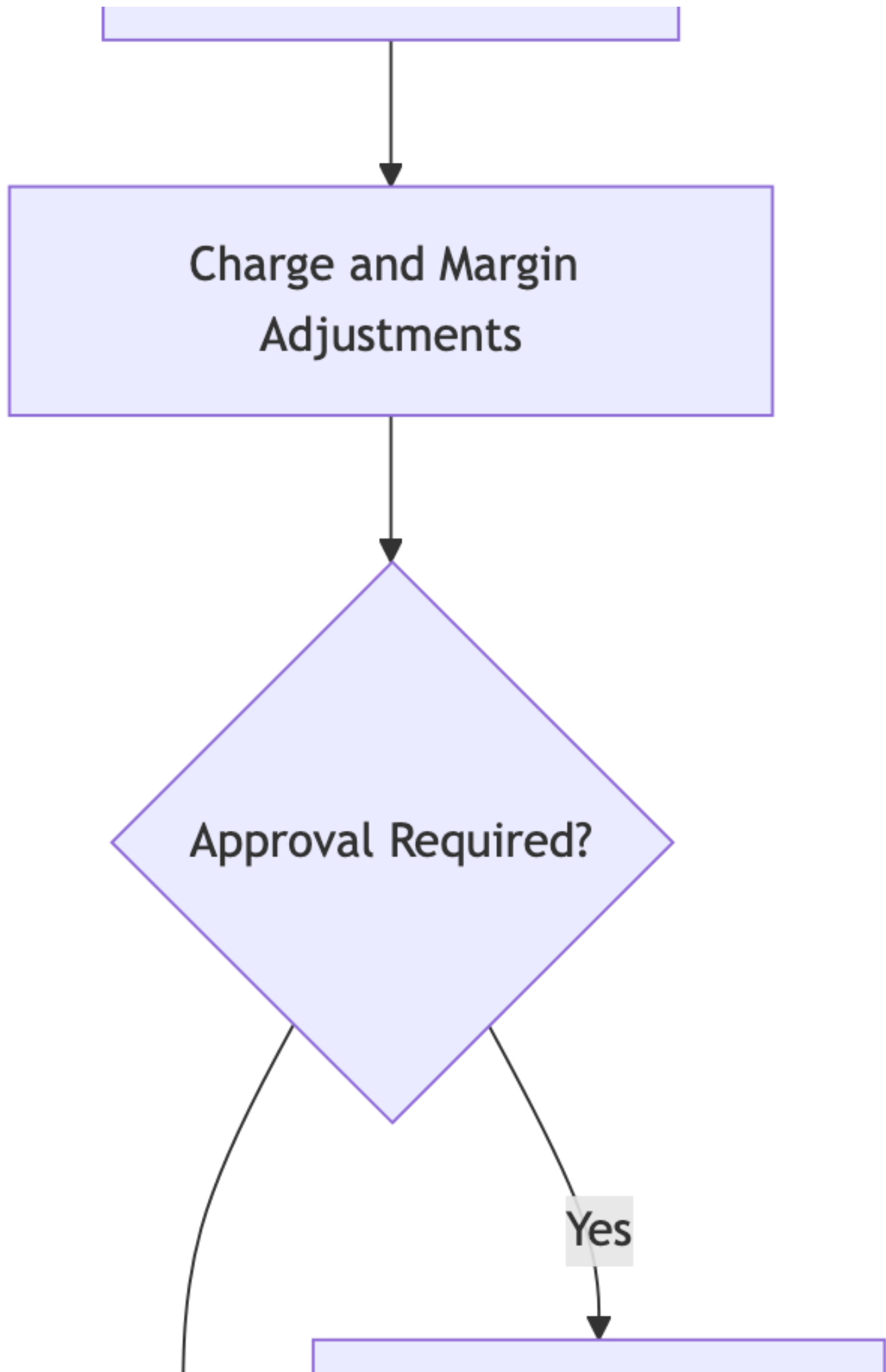
Composer Setup

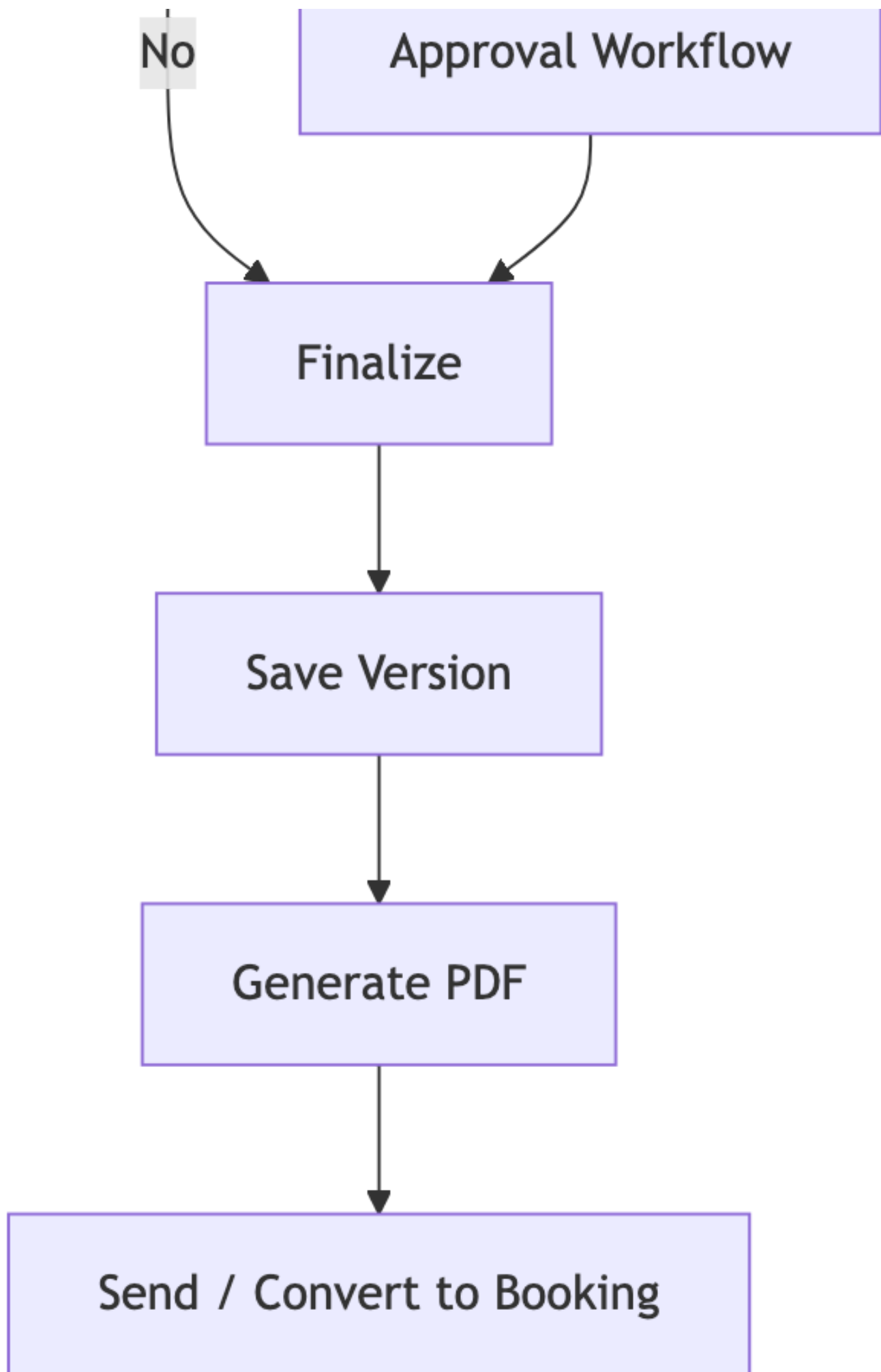


Option Generation



Option Comparison





12) Non-Functional Requirements (Cross-Cutting)

- Performance:
 - quote open p95 < 1.5s;
 - save quote p95 < 1.5s for standard payload;
 - compare options render < 250ms for 20-option set.
 - Reliability:
 - transactional integrity for quote+option+leg+charge save;
 - recoverable failures with user-safe retries.
 - Security:
 - strict tenant/franchise isolation;
 - RBAC enforcement for template/config/export-sensitive paths.
 - Observability:
 - structured logs keyed by quote_id, version_id, tenant_id;
 - error categories for schema, RPC, timeout, integration.
 - Backward compatibility:
 - additive schema changes only;
 - versioned API behavior for contract changes.
-

13) Implementation Readiness Checklist

- ☐ Consolidate field dictionary from schema + migration deltas into machine-readable artifact.
 - ☐ Define collaboration lock model and conflict policy.
 - ☐ Add composer drag-and-drop interaction contract and keyboard parity.
 - ☐ Finalize approval UX and integration hooks.
 - ☐ Formalize performance/error SLO dashboards.
 - ☐ Complete visual prototype package with all required annotated states.
 - ☐ Execute regression suite (unit/integration/e2e) and attach evidence.
-

14) Conclusion

The Quotation module already provides a strong operational baseline with robust data structures, multimodal routing, financial logic, and version controls. The highest-value next steps are collaboration, interaction maturity (drag-and-drop), governance unification (field registry + approval UX), and scalable observability. This PRD provides the complete reference needed for execution planning, architecture review, and phased delivery.